



Boiler OFF (Cut Out) at 75 psi
Boiler ON (Cut In) at 55 psi
Boiler maintains (Setpoint) 60 psi via PID

NOTE: NEW I (integral time) and D (derivative time) values - Option Parameters 21.3(22.3) and 21.4(22.4), may be required to achieve desired results.

Absolute Example – setting Opt 21.5 to 1:

Opt 21.1 (22.1)*set to..... 100 psi
Opt 21.5set to..... 1
Opt 21.6 (22.6)*set to..... 115
Opt 21.7 (22.7)*set to..... 95

RESULTS:

Boiler OFF (Cut Out) at 115 psi
Boiler ON (Cut In) at 95 psi
Boiler maintains (Setpoint) 100 psi via PID

Changing ONLY the Setpoint –

Opt 21.1 (22.2)change to..... 60 psi

NEW RESULTS from Setpoint Change:

Boiler OFF (Cut Out) at 115 psi
Boiler ON (Cut In) at 95 psi
Boiler maintains (Setpoint) 60 psi via PID

NOTE: NEW Cut In and Cut Out values as well as new I (integral time) and D (derivative time) values - Option Parameters 21.3(22.3) and 21.4(22.4), may be required to achieve desired results.

In BOTH examples (*) represents PID2

9.7.5 Manual Modulation

Pressing **AUTO/MAN** key toggles the burner in and out of manual or automatic modulation. To place the boiler in manual from automatic no password is required.

1. Press the **AUTO/MAN** key
2. Use the up/down arrow keys to set the firing rate.
3. Press the **AUTO/MAN** key again to place in automatic

Note: The firing rate will be shown on the display. This function can be disabled via option parameter 1.1. To do so requires the suppliers (highest) passcode.

9.7.6 Resetting the Control



To reset the Nexus, the **MUTE/RESET** key will have to be pressed and held for three seconds to first silence (mute) the alarm and then a second press for three seconds to clear the fault. A prolonged key press will do both.

If for some reason the Nexus detects a fault, the cause of the fault will be displayed if still present. If not displayed, pressing the **ENGINEERS** key once will reveal the fault, a second press will reveal the fault and subset in the upper right hand corner. The **ENGINEERS** key is hidden behind the Fireye® logo on the display. Pressing the logo will show the EK # to the right of which will be the fault # and any subset. Using the up/down arrow keys scrolls through the EK #'s to view other information. For specific Engineer's Key listing see Section 6.6 of Fireye Bulletin PPC6001.

9.7.7 Fault and Fault Finding

Should the Nexus determine a fault has occurred that required the burner be shut down, the bottom portion of the display will indicate the cause. Pressing the **ENGINEERS** key will display more information, which will be found in the upper right of the display. See Section 6.0 of Fireye Bulletin PPC-6001 for more details. Using the **ENGINEERS** key the last 33 faults or limits can be reviewed. The fault history begins with the most recent which can be found after **EK200** and will be date and time stamped. There are three aspects to each record; when it was reset, when it changed from a fault to a cleared fault and when the fault occurred. The portion of the record at the "**Fxx**" will also include the subset if any. **See Fault Subsets.**

Many of the faults detected and displayed by the unit will cause a safety shutdown of the burner. Other faults will be displayed as a warning, but operation of the burner will not be affected.

All fault numbers are prefixed by a letter. This letter describes the status of the fault:

- F - This means that a fault (e.g. PSU calibration fault) has occurred and is still present. The fault may be internal or external to the unit.
- L - This means that an alarm limit fault (e.g. smoke level exceeding the high limit) has occurred and is still present.
- C - This means that a fault or alarm limit did occur but has been cleared.

For example:

- F60 - Means fault 60 has occurred and still exists.
- L75 - Means alarm limit 75 has been exceeded and is still being exceeded
- C73 - Means fault or alarm limit 73 has occurred but has subsequently been cleared.

When the unit is in a safety shutdown state both the controlled shutdown triac and safety shutdown relay will be turned off, to prevent the burner from firing.

9.7.8 Fault subsets

As an aid to fault finding, most faults have a *fault subset* that gives additional information about the type of fault or what the burner was doing when the fault occurred. Where applicable, the subsets are given in the fault listing in section 0 of Fireye Bulletin PPC6001. For details of how to view a subset, refer to Section 6.0 of Fireye Bulletin PPC6001.



9.7.9 Fault listing

For specific Fault listing see Section 6.5 of Fireeye Bulletin PPC-6001.

9.7.10 PID Tutorial

PID Adjustment

The PPC6000 utilizes an advanced algorithm in order to maintain setpoint over a variety of load conditions. This three term PID can be infinitely adjusted to suit almost any application. The operator should have a basic understanding of the relationship between the three terms; proportional, integral and derivative.

Proportional

Typical older modulating systems employ only proportional control. This would be similar to the slide wire type found on most steam boilers. When using only a proportional control the system rarely achieves setpoint as the burner firing rate is lowered as the pressure comes up. At some point the input meets the actual demand and the pressure no longer raises or lowers, thus an offset between desired setpoint and actual operating pressure occurs. The only time the pressure and setpoint are the same is if the actual load equals the lowest firing rate of the burner, this is rare. An example of proportional only set up might be; Setpoint is 100 PSI, proportional range is 10 PSI. That is low fire is at 100 PSI, high fire is at 90 PSI with a 1:1 relationship in between, e.g.: 95PSI equals 50% rate. A good starting point for the "P" value, in most cases is about 10% of setpoint.

Integral

If the integral term is turned on, the control compares the actual pressure against setpoint at an adjustable interval. If there is an offset, the firing rate is increased by a small percentage until the next interval. This will continue until the pressure equals the setpoint. The same routine occurs as the pressure rises above the setpoint. Too much or too little integral will cause over and undershoot of the setpoint. Integral is set in seconds per repeat in the PPC6000.

For example, if the P were set at 10 psi with a boiler setpoint of 100 psi, the burner would first remain at high fire until the pressure reached 90 psi, and then start to modulate down. If the load was equal to 50% firing rate, the pressure would stall at 95 psi. Adding an "I" value of 10 would ramp the output up to 100% (high fire) in 10 seconds, assuming no change in the boiler pressure. Setting an "I" value of 300 increases the output up to 100% (high fire) in 5 minutes (300 seconds). As the boiler reaches setpoint, the same timing effect occurs above setpoint. Therefore, too much "I" can cause over/under shoot as loads change. Too little "I" may cause rapid hunting. A good starting point for the "I" value, in most cases is 50.

Derivative

Enabling the derivative has the effect of sensing the rate of change in the process variable (pressure or temperature) and increasing or holding up the firing rate output despite the integral term. This in effect amplifies the output to anticipate the effect of a sudden change in load demand. Derivative acts inversely when the setpoint is exceeded. Derivative is set in seconds on the PPC6000. A good starting point for the "D" value, in most cases is 2.

A rule of thumb is; start with a P value of about 10% of setpoint, an I value of 50 seconds and a D value of 2 seconds. After observing the operation through normal load swings, adjust each value, usually one at a time, and observe the results. When making a change it is recommended the value be doubled or halved to determine which direction you need to go. Bear in mind, the burner should not continuously hunt or swing to



maintain the desired setpoint. It is normal for a slight over and undershoots of setpoint during serious load changes.

The values for the PID's are found at option parameters 21.2, 21.3 and 21.4 for setpoint 1 (PID1), and 22.2, 22.3 and 22.4 for setpoint 2 (PID2). These values are adjustable using the **Site Passcode 154**.

9.7.11 The startup / shutdown stages are as follows:

Stage no.	Stage name	Description
0.	Non-volatile lockout / safety shutdown	The burner is held in this state until all faults are removed. The 'safety shutdown' output (PE5 – PE6) will be open. The 'controlled shutdown' output on PE3 will also be OFF during this time, however if the burner was running when the fault occurred, the 'safety shutdown' output will have opened first.
1.	Burner off (EK18)	The burner is checked to make sure that it has switched off completely. The fuel/profile select inputs must all go OFF when the burner is switched off (or at the end of post-purge). This provides a feedback to confirm that the burner is off, and ensures that this control is always synchronized with the burner controller. The 'controlled shutdown' output on PE3 will be OFF during this time, however the 'safety shutdown' output PE5 – PE6 will be closed during this time. The control will advance to status 2 when: - All fuel/profile select inputs are OFF - The 'boiler status' is equal to 1 (call for heat). See engineers key 18. - The 'AUTO' input (PA10 – PA11) is OFF (open).
2.	Wait for purge (EK5) (EK11-14) (EK31)	The 'controlled shutdown' output on PE3 will be switched on, to tell the burner controller there is a call for heat. This control waits for a fuel/profile select signal on one of the (line voltage) terminals PE9, PE10, PE11 or PE12 and a purge request (HIGH input, PA9 - PA11 closed. This input is low voltage). This would normally come from the burner controller. If the fuel/profile select input is removed (PE9, 10, 11 or 12), the control will immediately move to status 1.

Stage no.	Stage name	Description
3. & 4.	N/A	Not Applicable to PPC6000 Note: Statuses 3 and 4 are reserved for a gas valve proving system and are not implemented on this control. This control will advance directly to status 5 once the conditions above have been met.
5.	Prove closed positions (EK80-89)	The fuel and air motors are moved down until they are stopped by the 'closed position' micro-switches in the servos. The final positions are compared with the closed positions stored in memory, and must be within $\pm 5^\circ$ of the stored values otherwise the control will lockout. Variable speed drives must read zero. (4 mA) When all drives have stopped moving, the control will advance to status 7. The 'HIGH' input (PA9 to PA11) still must be made during this time. If the fuel/profile select input is removed (PE9, 10, 11 or 12), the control will immediately move to status 1.
6.	N/A	Not Applicable to PPC6000 Note: Status 6 is reserved for air pressure detection, and is not implemented in this control. This function must be provided by the burner controller.
7.	Moving to Purge	The selected motors are moved up towards the purge position. When the drives have all stopped, the control moves to status 8. If the fuel/profile select input is removed (PE9, 10, 11 or 12), the control will immediately move to status 1.
8.	Pre-purge	The controller confirms all required drives are at their purge positions, and gives a 'purge proved' signal by providing (line voltage) to terminal PE8 (high-fire-switch). The control will remain in this status until the burner controller signals the end of pre-purge by breaking PA11 – PA9 ('HIGH' input). If the fuel/profile select input is removed (PE9, 10, 11 or 12), the control will immediately move to status 1.
9.	Move to ignition positions	The selected motors are moved to their ignition positions. The 'purge proved' signal is switched off. When the drives have all stopped moving, the control advances to status 10. If the fuel/profile select input is removed (PE9, 10, 11 or 12), the control will immediately move to status 1.



Stage no.	Stage name	Description
10.	Ignition	The controller confirms all required drives are at their ignition positions, and gives an 'ignition proved' signal by supplying line voltage to terminal PE7 (low-fire-switch). The controller will hold the drives at their respective ignition positions until the signal to modulate is received from the burner controller. This is performed by closing the circuit on PA10 – PA11 (AUTO input). If the fuel/profile select input is removed (PE9, 10, 11 or 12), the control will immediately move to status 1.
11-14	N/A	NOTE: Statuses 11 to 14 control the light-up sequence of the burner and are provided by the 'burner controller' device used for the application. This control will jump from status 10 to status 15 when the light up is completed. Not Applicable to PPC6000
15.	Moving to low fire	Once the AUTO signal is received the fuel and air motors are moved from their ignition positions to their low fire positions. These positions may or may not be the same as the ignition positions. The 'ignition prove' terminal PE7 (low-fire-switch) is switched off. If the fuel/profile select input is removed (PE9, 10, 11 or 12), the control will immediately move to status 1. Once the drives have reached their low fire positions, and an AUTO signal is received (PA10 - PA11), the control will advance to status 16. IF Option 23.0 (warming limit) is enabled, drive will remain at low fire until time set by Option 23.0 expires. NOTE: terminal PE7 will always be OFF during status 15.

Stage no.	Stage name	Description
16.	Modulation (EK33)	Once the fuel and air motors reach their low fire positions, they are modulated according to the demand placed on the burner. In this burner status the 'high-fire-switch' and 'low-fire-switch' outputs (PE8 & PE7) will come on at high and low fire respectively and may be used for indication purposes. If a 'HIGH' input is provided by closing PA9 – PA11, this control will move to status 17. If the 'AUTO' input is removed by opening PA10 – PA11, this control will modulate down to low fire then jump to status 15. Note: Terminal PE7 will not be energized in this case. This may be used to provide a low fire post purge, or a low fire hold function. If the fuel/profile select input is removed (PE9, 10, 11 or 12), the control will immediately move to status 1. If this control wishes to turn the burner off (there is no call for heat, for example), it will open the 'controlled shutdown' relay, removing power from PE3. It will remain in modulation status however, until one of the conditions above is met.
17.	Move to post-purge	The selected motor(s) are moved to their purge positions all others are moved to their closed positions. When the drives have all stopped, the control will move to status 18.
18.	Post-purge	The controller confirms all required drives are at their purge positions, and gives a 'purge proved' signal by providing (line voltage) to terminal PE8 (high-fire-switch). The control will remain in this status until the burner controller signals the end of pre-purge by breaking PA9 – PA11 ('HIGH' input), or the fuel/profile select input is removed (PE9, 10, 11 or 12). In either case, the control will immediately move to status 1.

9.7.12 Non-volatile lockout

Non Volatile lockouts cannot be cleared without operator intervention and are remembered in the event of power being removed from the control.

A non-volatile lockout will occur under the following conditions:

- In any stage the interface signals are incorrect.
- In stages 5, 7 and 8, stages 10-16 (inclusive) and stage 18 if a motor is not in the correct position
- In any stage, if an internal or external fault not previously mentioned occurs which may affect the safe operation of the burner (see section 0 of Fireeye Bulletin PPC6001)



9.8 Fireeye Nexus Combustion Curve

Site: _____

Date: _____

Profile: _____

Fuel: _____

NAM E	Drive 0	Drive 1	Drive 2	Drive 3	Drive 4	Drive 5	Drive 6	Drive 7	Drive 8	Drive 9	O2	CO	NOx
P 0													
P 1													
P 2													
P 3													
P 4													
P 5													
P 6													
P 7													
P 8													
P 9													
P 10													
P 11													
P 12													
P 13													
P 14													
P 15													
P 16													
P 17													
P 18													
P 19													
P 20													
P 21													
P 22													
P23													

Notes: PID Setup: P = I = D =

10 Function Block Programming Details

Below is an *overview* of the capabilities of the function block program – Abacus. This overview is intended to give the user insight into what can be accomplished with function block programming.

NOTE: For a more complete description on Abacus, see Fireye Bulletin NXAB-1001.

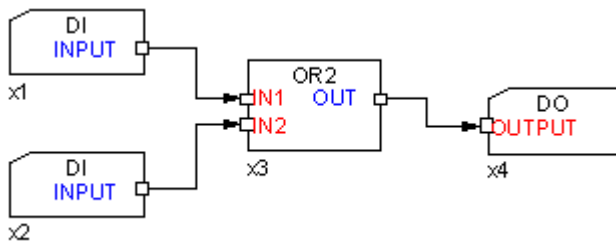
Introduction to Fireye's Abacus Programming Software

1.1. Introduction

Abacus is a powerful software package that is used to program Fireye PPC6000 and NX6100 series controls. The software allows the user to define special functions such as draft or feed water control to be implemented within the control using standard or optional inputs and outputs. The designer can assign option parameters and engineer's key's as well as levels of password protection to the custom options. After the program is completed it is then uploaded to the control via optional Fireye Comfire software. The user can run the optional program by selection in option parameter 00.4. The factory default program is always available as a back up by deselecting the user program.

The programs are entered graphically by drawing and connecting blocks on schematic pages. These schematics are then sent to the controls where they operate as a program.

Here is a preview of a simple program.



Briefly, this program will give a relay closure (digital output, DO) if either of the digital inputs (DI-1 or DI-2) is activated. These inputs could come from relay closures on other equipment.

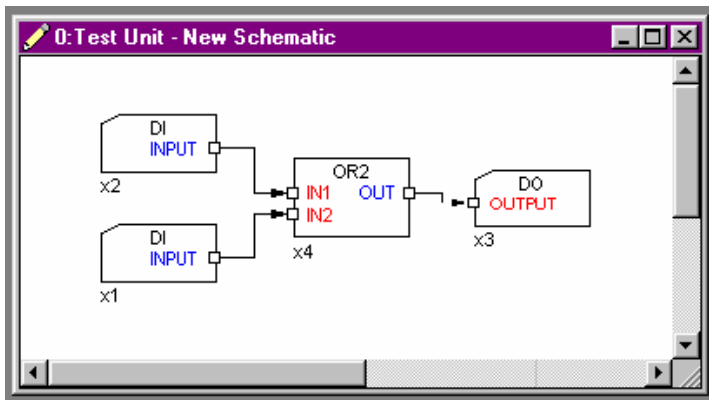
The details of which inputs are read, which output is activated are covered in Fireye Bulletin NXAB-1001.



WARNING: ON NO ACCOUNT MUST THE PROGRAMMABLE BLOCKS BE USED FOR ANY SAFETY CRITICAL FUNCTION.

The operation of the burner management / drive positioning system in the basic control is fail-safe however the programmable blocks system is not. If the programmable blocks are used to control a safety related item (such as water level) then appropriate external limit checking must be provided to maintain safety if the user program should fail to operate.

After loading and executing the Abacus software and following the required steps, a blank schematic page will appear. The toolbar on the right hand side of the main abacus window is used to place blocks on the screen.



First draw the blocks, then connect them.



Using the toolbar, click the top button **twice** to bring up two digital input blocks.

You will notice that the blocks always appear in the middle of the visible window. In this case, one is on top of the other. Move these to the left hand side of the screen by dragging with the mouse. To do this, you must click within the block area, not on one of the pins (square nodes).

You will also notice that each block has an x-number by its bottom left corner. This is simply a reference number for the block. Each block on a page will have a unique x-number.



On the toolbar, click the second button from the top **once**. This will give up a digital output block. Move this towards the right hand side of the screen.



On the toolbar, click the OR gate symbol **once** to create an OR function block. Move this to the middle of the other blocks, roughly as on the example.

To connect the blocks together:

- ☐ Move the mouse pointer over the output pin of the block the signal is to originate from.

The output pins are those which have a description shown in blue. They are always on the right hand side of a block. The term 'output pin' refers to the output from this block to the next one, not an output from the control itself.

- ☐ Press and hold the left mouse button.
- ☐ Move the mouse pointer over an 'input pin' of the block the signal is to arrive at, keeping the left mouse button pushed.
- ☐ Release the mouse button. A link should be drawn.

Links must always be drawn like this - starting from an output, finishing at an input.

Note: Some of the blocks have a sloped top-left corner and some do not. These blocks require parameters.